Structural Compatibility of the Standard Model with Induced Scalar-Pressure Gravity: a Partial Embedding and Extension Program

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Version 1.0 (Companion appendix to the MASS paper)

Abstract

We present a six-phase analysis of how the Standard Model (SM) gauge and fermion content can be hosted in the cavity framework of Induced Scalar-Pressure Gravity (ISPG). The scope is *compatibility*, not derivation from first principles: given the ISPG archion substrate and the Mathieu ladder previously established, we check whether the SM quantum numbers, anomaly conditions, and electroweak structure can be assigned consistently to cavity labels, and which elements of the SM must still be imported. We find: (i) the count of confined modes of the D_3-deformed oscillon is eight, matching $\dim \mathrm{SU}(3)_{\mathrm{adj}}$, without fine-tuning; the SU(3) Lie-algebra structure itself is not independently derived from the cavity dynamics in this work; (ii) the integer cavity labels N_W, N_Z individually fit m_W, m_Z to $\leq 0.31\%$; their ratio $(N_W/N_Z)^2 = 0.8817$ matches the on-shell mass-ratio convention $\cos\theta_W^{\mathrm{os}} \equiv m_W/m_Z \simeq 0.8815$ to 0.03%—an *internal consistency check*, not an independent prediction of the Weinberg angle, since the two indices are themselves fits to the masses whose ratio defines $\cos\theta_W^{\mathrm{os}}$ at SM tree level; (iii) the ISPG action contains no Φ^4 term, so a SM-type Higgs mechanism is ruled out at the fundamental level; the specific replacement dynamics for gauge-boson mass generation is not derived here and is listed as an extension item; (iv) the hypercharge pattern $Y \cdot N_c \in \mathbb{Z}$ is a compatibility and integer-grading clue respected by the imported SM assignment; it is not independently derived here, and the SM-specific hypercharge values are imported via the generation-1 fitting formula plus Gell-Mann–Nishijima; (v) the three generations are interpreted (under a model-dependent three-axis hypothesis) as the three orthogonal vibration axes of the 3D archion, giving a geometric reading of the number 3; (vi) neutrinos lie *outside* the Mathieu-ladder fermion formula and are taken as polarization-rotation modes per the companion MASS paper; their inclusion in the formula would mispredict m_ν by six orders of magnitude, and the exclusion is therefore structural, not cosmetic. Extension topics (the 3D vector cavity spectrum behind the individual indices N_W, N_Z; the SU(3) Lie-algebra structure beyond the count; the archion-interior mechanism for gauge-boson mass generation; the fine-structure constant; CKM and PMNS matrices; the exact charged-lepton mass hierarchy) are identified as a programmatic continuation and do not conflict with the compatibility established here.

Contents

1 Introduction and scope

The ISPG framework [45, 47] postulates a single real scalar field $\Phi(x)$ coupled minimally to gravity through a pressure term. Earlier ISPG papers [45, 47] establish: (a) the bare action contains only the curvature R and the kinetic term $\frac{1}{2}(\partial\Phi)^2$, with *no* non-derivative potential; (b) localized finite-energy solutions ("archions") exist as time-periodic oscillons inside a finite cavity; (c) the linear spectrum of small fluctuations around the archion obeys Mathieu's equation, whose characteristic values $b_N(q)$ at fixed $q \simeq 1.853$ reproduce all known charged elementary particle masses through $m_N = m_1\, b_N(q)$ to better than 0.3% [80–82].

Gap addressed. What the earlier ISPG papers did *not* settle was whether SM quantum numbers, anomaly conditions, and electroweak structure can be *hosted consistently* on the cavity labels that ISPG natively provides—and, if so, which SM ingredients are imported from external knowledge versus derived from the cavity. This paper carries out that compatibility check explicitly across six phases, with a clear separation (via epistemic labels) of what is derived, what is fit, what is imported, and what is deferred. The result is a *partial* structural embedding: sufficient to exclude gross inconsistencies and to identify the specific open items (§9) that a *complete* embedding would still need to resolve.

Methodological commitments. Throughout, each physical statement carries an explicit epistemic label: *[derived here]* (proved in this paper from the ISPG substrate), *[standard SM]* (used as a known SM fact), *[numerically verified]* (verified numerically by a Python script in `python/`), *[model-dependent]* (holds under a specific hypothesis, stated as such), or *[deferred to extension program]* (part of the extension program). The six phases A–F below partition the embedding into structurally separable tasks; each phase terminates in an explicit pass/fail criterion.

Non-goals. This paper does *not* derive the exact masses of the muon and tau, does *not* derive the Cabibbo–Kobayashi–Maskawa (CKM) or Pontecorvo–Maki–Nakagawa–Sakata (PMNS) mixing matrices, and does *not* reconstruct the fine-structure constant α from first principles. These are open targets for the extension program (§9). Their *absence* does not invalidate the structural embedding; it constrains the scope of what the embedding establishes.

2 ISPG prerequisites used below

The following ISPG results are taken as inputs to the embedding; each is proved in the referenced paper or Python verification script.

Action. *[from [45]]* The bare action is

$$S = \frac{1}{16\pi G}\int d^4x\,\sqrt{-g}\,\left[\,R + \tfrac{1}{2}(\partial\Phi)^2\,\right] + S_m, \tag{1}$$

with S_m the standard matter action. There is no Φ^2, Φ^4, or generic $V(\Phi)$ term; consequently any potential appearing below is *effective*, induced by nonlinear self-interaction through archion dynamics [90, 91], not added by hand.

Mathieu ladder. *[from [45]]* Linearized fluctuations around the periodic archion solve Mathieu's equation. At the characteristic deformation $q \simeq 1.853$, the characteristic values $b_N(q)$ form a discrete ladder indexed by $N = 1, 2, 3, \ldots$, and every massive SM particle sits on an integer N. The mass law

$$m_N = m_1 \cdot b_N(q) \tag{2}$$

is the master interpretation of N as a cavity-mode label. Empirical fits (for the electron taken as $N_e = 5$) give $m_N \propto N^2$ to high accuracy in the heavy sector; this quadratic scaling is used in phases D and F (`phase_d_ew_breaking.py`).

Archion and D_3. *[from [47]]* An archion with triangular (D_3) deformation in three spatial directions admits a tensor-mode spectrum whose confined sector contains exactly eight states for a robust range of the deformation parameter (documented in `SM/ISPG_EightGluon_Robustness.md`); this is used in §3.

Three-axis hypothesis. *[numerically verified] [model-dependent]* The 3D archion can sustain three orthogonal vibration modes (one per spatial axis), each with its own damping. Under this hypothesis, the (e, μ, τ) charged leptons are identified with one oscillator on three axes, and the Koide relation is reproduced to 0.0006%, Sargent's law to 0.26%, and the absolute τ mass to 0.4%. This hypothesis is model-dependent; the embedding in §7 uses it to *interpret* the number

3 of generations but does not need the exact numerical reproduction of the three charged-lepton masses.

Two forms of the mass law. The 0.0006% Koide figure quoted above is obtained with the full Mathieu characteristic values $a_N(q \simeq 1.853)$ from the linearized archion spectrum. The large-N approximation $m \propto N^2$, used elsewhere in this paper for the heavy sector (electroweak bosons, Higgs, and the cross-sector compatibility survey), is the asymptotic limit of the same ladder: it matches the observed masses to sub-percent precision for $N \gtrsim 70$ but returns $Q = \frac{1}{2}$ for Koide rather than the observed $Q \simeq \frac{2}{3}$. Precision-level lepton statements (Koide, Sargent) therefore use the full Mathieu evaluation, while compatibility-level embedding statements in §§3–7 use the N^2 form. No additional physical input is involved; the two forms agree asymptotically.

3 Phase A — Gauge sector

Script. `python/phase_a_gauge_sector.py`.

3.1 SU(3): eight confined gluons from D_3

[numerically verified] (octet-count clue; documented in the robustness analysis, supplementary notes on file with the author; summarized here). The D_3-deformed three-dimensional oscillon carries a tensor perturbation spectrum with ten modes at leading order; exactly eight are confined (lie below the chosen leak threshold) across a nonzero, threshold-sensitive region of parameter space

$$g_0 \in [0.1, 8.0], \qquad V \in [3.0, 9.5], \tag{3}$$

with the central value $V = \Delta \approx 6.26$ identified as the resonant deformation. The confined multiplicity is *discrete-binary* in $\{8, 10\}$: the values $\{5, 6, 7, 9\}$ never appear. This matches the dimension of the SU(3) adjoint representation (dim = 8) as a structural compatibility clue. The robustness analysis gives an $N = 8$ region of 13.23% at the default 40% leak threshold, 6.50% at 45%, and 0% at 50%; the physical threshold and the parameters g_0, V must still be derived from the full $G_2(X)$ dynamics.

Caveat: count versus algebra. *[deferred to extension program]* The count of eight confined modes is a necessary but not sufficient condition for an SU(3) gauge structure. The full identification requires that the eight modes satisfy the Gell-Mann commutation relations $[T^a, T^b] = if^{abc}T^c$ with the standard SU(3) structure constants f^{abc}. Extracting these constants from the nonlinear archion dynamics (specifically, from the trilinear mode-coupling vertices) is a separate computational task deferred to item E6 of the extension program. In its current form, the result of this subsection is therefore "$\dim \mathrm{SU}(3)_{\mathrm{adj}}$ is natively reproduced," not "SU(3) is derived."

3.2 SU(2) × U(1)$_Y$: numerical consistency

[numerically verified] The electroweak bosons sit on specific Mathieu indices:

Particle	m (GeV)	N	$m_{\mathrm{pred}} = (N/5)^2 m_e$	error
$W^{\pm}$	80.369	1986	80.62 GeV	0.31%
Z^0	91.188	2115	91.44 GeV	0.27%
H^0	125.250	2479	125.61 GeV	0.29%

This table uses the rough large-N estimate $m_N = (N/5)^2 m_e$. The full Mathieu normalization gives a separate precision diagnostic ($W \simeq 80.389$ GeV, $Z \simeq 91.172$ GeV, $H \simeq 125.254$ GeV), but it still does not derive the labels (N_W, N_Z, N_H). *[numerically verified]* The ratio

$$\left(\frac{N_W}{N_Z}\right)^2 \;=\; \left(\frac{1986}{2115}\right)^2 \;=\; 0.8817, \tag{4}$$

is to be compared with the on-shell mass-ratio convention $\cos\theta_W^{\rm os} \equiv m_W/m_Z \simeq 0.8815$ [82] and with the observed $m_W/m_Z = 0.8814$: a $\leq 0.03\%$ agreement in each case.

Nature of this test: consistency, not independent prediction. Since N_W and N_Z were themselves fit to m_W and m_Z individually (to 0.31% and 0.27% respectively), their ratio *automatically* matches m_W/m_Z within the combined fit residuals. The ratio test therefore does not constitute an independent prediction of the Weinberg angle; at SM tree level in the on-shell convention $m_W/m_Z = \cos\theta_W^{\rm os}$ holds identically and any theory that reproduces the two masses to high accuracy will reproduce this ratio. The non-trivial content of (4) is that the *discrete integer* Mathieu ladder (step size $\Delta N = 1$ out of $N \sim 2000$) can simultaneously host both bosons on the same mass law $m \propto N^2$ with residuals well below one ladder step; the ratio inherits this precision mechanically.

What is open. *[deferred to extension program]* The first-principles derivation of the individual labels (N_W, N_Z) from the three-dimensional vector cavity spectrum is the principal outstanding task for a non-trivial electroweak prediction (item E1). Until it is completed, the electroweak sector of the present embedding is a consistency check of the Mathieu ansatz against two data points, not a structural derivation of the Weinberg angle.

3.3 Phase A verdict

Yellow on the gluon count: the D_3-deformed oscillon gives an octet-count compatibility clue, with partial numerical robustness and threshold sensitivity, while the full Lie-algebra structure is deferred (E6). Yellow on $\mathrm{SU}(2)\times\mathrm{U}(1)_Y$: the Mathieu ladder accommodates W, Z, H with sub-percent residuals, but the electroweak test is internal consistency rather than an independent prediction of the Weinberg angle. The derivation of the specific labels (N_W, N_Z) is deferred (E1). No contradiction with ISPG's single-scalar action is introduced by either step.

4 Phase B/C — Fermion quantum numbers

Scripts. `python/phase_c_quantum_number_map.py`, `python/phase_c_step2_T3_Y_map.py`.

4.1 Generation-1 SM-to-cavity compatibility map

[numerically verified] We test whether the imported SM labels (Q, N_c) can be used to assign cavity-compatible generation-1 Mathieu indices for the massive charged fermions (e, u, d). An ansatz using these imported labels

$$N \;=\; 5\big[\,1 + N_c\big(1 - |Q|\big)\big] \tag{5}$$

reproduces the three observed indices. The implied structure is

$$3|Q| \;\in\; \{0, 1, 2, 3\}, \tag{6}$$

i.e. Q takes only values whose triple is an integer: this is a third-integer charge grading. Interpreting this grading as a $\mathbb{Z}_3$ or D_3 representation requires a separate representation map and is left open.

Status: fit, not prediction. The formula (5) has three free ingredients (the anchor $5 = N_e$, the coefficient 1 multiplying N_c, and the choice of functional form $1 - |Q|$) and is fit to three data points (N_e, N_u, N_d). A perfect match is therefore guaranteed by parameter counting and is not by itself predictive. The structural content that *does* survive parameter counting is the *shape*: that a formula built from the imported SM labels (Q, N_c), anchored on the electron and returning cavity-compatible integer multiples of 5, exists at all. A genuine prediction from (5) would require either (i) a fourth data point that the formula was not fit to, or (ii) an independent derivation of the coefficients from the archion dynamics. Both are deferred.

4.2 Neutrinos are excluded from the Gen-1 formula

[derived here] Formally, substituting $Q = 0$, $N_c = 1$ into (5) gives $N_{\nu_e} = 10$, which via the $m \propto N^2$ scaling would imply $m_{\nu_e} \approx (10/5)^2 m_e = 2.044\,\mathrm{MeV}$. This exceeds the observed neutrino mass bound $m_\nu < 1\,\mathrm{eV}$ by more than six orders of magnitude. The formula therefore *cannot* apply to neutrinos.

The resolution is structural, not ad hoc: the companion MASS paper [45] identifies neutrinos as *polarization-rotation* (helical-torsional) modes of Φ, distinct from the Mathieu-ladder oscillon family, with their tiny mass arising from the suppressed torsional rigidity of the nonlinear $G_2(X)$ coupling. Under that identification, neutrinos are not cavity oscillons at all and the Mathieu mass law $m \propto N^2$ does not apply to them. Consequently the scope of (5) is explicitly restricted to the *massive charged* fermion sector, and the nominal $N = 10$ assignment for ν_e is discarded.

This exclusion eliminates a potential internal contradiction of the embedding but, equally, limits its reach: the Gen-1 formula says nothing about neutrino quantum numbers, and the T_3, Y assignments for ν_L in Table 1 are imported from the SM via Gell-Mann–Nishijima, not derived from the cavity.

4.3 Hypercharge map and Gell-Mann–Nishijima

[derived here] The full left-Weyl content of generation 1 is listed in Table 1. For every entry, the Gell-Mann–Nishijima relation

$$Q = T_3 + \tfrac{1}{2}Y \tag{7}$$

is satisfied as an identity.

Table 1: Generation-1 Weyl-fermion content (left-handed basis, right-handed fields written as charge-conjugated left-handed spinors).

Field	Q	T_3	Y	N_c	$Y \cdot N_c$	$Q = T_3 + Y/2$
ν_{eL}	0	$+\frac{1}{2}$	-1	1	-1	OK
e_L	-1	$-\frac{1}{2}$	-1	1	-1	OK
e^c_L	$+1$	0	$+2$	1	$+2$	OK
u_L	$+\frac{2}{3}$	$+\frac{1}{2}$	$+\frac{1}{3}$	3	$+1$	OK
d_L	$-\frac{1}{3}$	$-\frac{1}{2}$	$+\frac{1}{3}$	3	$+1$	OK
u^c_L	$-\frac{2}{3}$	0	$-\frac{4}{3}$	3	-4	OK
d^c_L	$+\frac{1}{3}$	0	$+\frac{2}{3}$	3	$+2$	OK

Integer hypercharge-times-color. *[standard SM] [numerically verified]* A direct reading of the last-but-one column gives

$$Y \cdot N_c \in \{-4, -2, -1, +1, +2\} \subset \mathbb{Z} \tag{8}$$

for every field (including both chiralities). This integrality is an observed compatibility clue: the imported SM assignment respects an integer grading when hypercharge is multiplied by color multiplicity. It is not, by itself, a derivation of the hypercharges or of anomaly cancellation. The anomaly equations require the specific set $\{-4, -2, -1, +1, +2\}$ with the specific multiplicities shown, not merely arbitrary integers. In this work the Y values are imported through the Gen-1 formula (5) combined with Gell-Mann–Nishijima; the product $Y\,N_c$ is therefore a checked compatibility pattern, not an automatic consequence of independently derived cavity labels. The integrality is also a hallmark of Pati–Salam-type unifications (J. C. Pati and A. Salam, *Phys. Rev. D* **10**, 275 (1974)), reached here through a different route.

Chirality. *[standard SM] [deferred to extension program]* The distinction (L/R) in the fermion content is imported from the standard Dirac structure of S_m; deriving chirality from cavity boundary conditions is a separate open problem (§9).

4.4 Phase B/C verdict

For the massive charged generation-1 fermions (e, u, d) the indices (N_e, N_u, N_d) are accommodated by a three-ingredient fit formula (5) built from imported SM labels (Q, N_c); the fit is exact on the three data points by parameter counting and is not yet predictive. Neutrinos are excluded from the formula by construction (§4.2). Gell-Mann–Nishijima holds exactly for every entry in Table 1 because the T_3, Y assignments are imported from the SM consistent with it. The integer structure $Y\,N_c \in \mathbb{Z}$ is a checked compatibility pattern respected by the imported SM assignment.

5 Phase D — Electroweak symmetry breaking without Higgs

Script. `python/phase_d_ew_breaking.py`.

5.1 No Φ^4 in the action

[derived here] The bare action (1) contains no Φ^2 and no Φ^4 term. A SM-type Higgs mechanism is therefore absent at the fundamental level: there is no $V_{\text{Higgs}}(\Phi) = \mu^2\Phi^2 + \lambda\Phi^4$ with $\mu^2 < 0$, no SU(2)-doublet scalar, and no vacuum expectation value $\langle\Phi\rangle = v/\sqrt{2}$ filling the universe homogeneously. This is a structural consequence of the single-scalar postulate and forces ISPG to generate gauge-boson masses by a different mechanism (see below).

5.2 Local EW breaking: the archion interior

[model-dependent] The role that the Higgs plays in the SM—providing a region of space where $\Phi \neq 0$ so that W, Z acquire mass— is played in ISPG, under this proposal, by the *interior of each archion.* An archion is a localized finite-energy excitation with $\Phi(r) \neq 0$ inside its core and $\Phi(r) \to 0$ outside. The SM's "broken phase" maps to the archion interior; the SM's "unbroken phase" maps to the laboratory vacuum. Hence EW symmetry breaking in ISPG is proposed to be *local*, not global.

What is not shown. *[deferred to extension program]* The specific dynamical mechanism by which gauge bosons acquire their observed masses from the archion interior—the analogue of the Higgs Yukawa couplings and gauge-kinetic term $(D_\mu\Phi)^\dagger(D^\mu\Phi)$—is *not* derived in this paper. A complete replacement would need to show how an SU(2) gauge field minimally coupled to the archion acquires a mass matrix matching M_W, M_Z (with the photon remaining massless) from the archion's spatial profile alone. Until this is done, the statement "ISPG does not need a Higgs" should be read as: "the ISPG action forbids a SM-type fundamental Higgs scalar by construction, and an alternative local mechanism is proposed but not derived." This task is item E7 in the extension program.

Two immediate consequences:

(i) The laboratory vacuum is genuinely empty of any Higgs condensate, consistent with the observed cosmological constant problem being an order-of-magnitude mismatch between SM-expected vacuum energy and observation.

(ii) The apparent scale $v \approx 246\,\mathrm{GeV}$ is an *internal length scale* of the archion, not a property of space at large. Via (2) this length maps to a large Mathieu index ($N \approx 3471$ in the $m \propto N^2$ extrapolation); it is not itself a particle.

5.3 Consistency check: $m_W/m_Z = \cos\theta_W^{\mathrm{os}}$

[numerically verified] As discussed in §3.2, the ratio (4) matches the on-shell mass-ratio convention $\cos\theta_W^{\mathrm{os}}$ to 0.03%. This is a self-consistency check of the Mathieu ansatz (the same integer ladder hosts both bosons), not an independent prediction of the Weinberg angle, since the individual indices N_W, N_Z are themselves fits to m_W, m_Z. What is genuinely non-trivial is that a *discrete* ladder can host two particles of such close but distinct masses simultaneously with sub-percent residuals: this constrains the ladder spacing to be fine enough at large N, which the $m \propto N^2$ law satisfies automatically.

5.4 Phase D verdict

The ISPG action (1) forbids a SM-type fundamental Higgs scalar by construction (no Φ^2, no Φ^4). A local alternative (gauge-boson mass generation inside the archion) is proposed but not derived (E7). The m_W/m_Z ratio consistency is a fit diagnostic, not a derivation of the Weinberg angle. No internal contradiction with ISPG is introduced, but the electroweak sector remains structurally incomplete until E1 and E7 are carried out.

6 Phase E — Anomaly cancellation

Script. `python/phase_e_anomalies.py` (exact arithmetic with `fractions.Fraction`).

6.1 The five conditions

[standard SM] A consistent gauge theory with the SM matter content must satisfy five independent anomaly-cancellation conditions:

1. $\mathrm{U}(1)_Y^3$: $\sum_L Y^3 = 0$;
2. $\mathrm{SU}(2)^2 \times \mathrm{U}(1)_Y$: $\sum_{\text{doublets}} Y = 0$;
3. $\mathrm{SU}(3)^2 \times \mathrm{U}(1)_Y$: $\sum_{\text{colored}} Y = 0$ (summed per SU(2) component);
4. gravitational $\times\,\mathrm{U}(1)_Y$: $\sum_L Y = 0$;
5. Witten SU(2): the total number of SU(2) doublets (counting color multiplicity) is even.

Here $\sum_L$ denotes a sum over all left-handed Weyl fermions (right-handed fields are charge-conjugated into their left-handed equivalents).

6.2 Evaluation on ISPG generation-1 content

Using Table 1 with the left-Weyl conjugate basis $\{L_L,\ Q_L,\ e^c_L,\ u^c_L,\ d^c_L\}$ (doublets counted once per SU(2) state, colored fields weighted by N_c):

#	Anomaly	Value	Status
1	$\mathrm{U}(1)^3_Y$	0	pass
2	$\mathrm{SU}(2)^2\mathrm{U}(1)_Y$	0	pass
3	$\mathrm{SU}(3)^2\mathrm{U}(1)_Y$	0	pass
4	grav $\times$ $\mathrm{U}(1)_Y$	0	pass
5	Witten SU(2)	4 doublets (even)	pass

All five anomalies cancel exactly (verified with rational arithmetic in the script).

6.3 What this tells us and what it does not

[standard SM] [numerically verified] The anomaly conditions are exact consistency checks on the imported left-Weyl SM hypercharge assignment. With the SM representation content and the standard hypercharge normalization fixed, changing an individual entry in $(Y_{L_L}, Y_{Q_L}, Y_{e^c}, Y_{u^c}, Y_{d^c})$ generically reactivates at least one anomaly. The specific SM values $(Y_{L_L}, Y_{Q_L}, Y_{e^c}, Y_{u^c}, Y_{d^c}) = (-1, +\frac{1}{3}, +2, -\frac{4}{3}, +\frac{2}{3})$ are imported through the Gen-1 formula (5) combined with Gell-Mann–Nishijima; they are not independently derived from archion dynamics in this paper.

The integer pattern (8) is therefore a parallel charge-quantization / integer-grading clue, not the theorem that cancels anomalies. The five anomaly equations above are what cancel for the imported assignment. In particular, the anomaly sums are homogeneous in the hypercharges, so a common rescaling can preserve cancellation while changing the integer normalization of $Y\,N_c$; conversely, an arbitrary integer $Y\,N_c$ pattern need not satisfy the five sums. What Phase E therefore establishes is: the SM hypercharge assignment, once imported onto the ISPG generation-1 content through the Gen-1 map, passes all five standard anomaly tests. In the Pati–Salam unification it is known that Y and N_c are linked so that $Y\,N_c$ is integer; ISPG sees the same integrality as a compatibility clue, without postulating a larger unifying gauge group.

6.4 Phase E verdict

All five standard anomaly conditions are satisfied by the imported SM hypercharge assignment on the ISPG generation-1 content. This is a consistency check of the imported charges against the cavity's integer grading; it does not on its own constitute a derivation of the SM hypercharges from archion dynamics.

7 Phase F — Generation replication

Script. `python/phase_f_generation_replication.py`.

7.1 Structural identity of the three generations

[standard SM] The three SM generations carry identical (Q, T_3, Y, N_c) assignments; they differ only by mass and by entries of the CKM and PMNS mixing matrices. The embedding therefore trivially extends to generations 2 and 3 at the level of quantum numbers.

7.2 Anomaly cancellation for $n_{\text{gen}} \geq 2$

[derived here] Each of the five anomaly conditions is linear in the generation count: if $A_i^{(1)} = 0$ on generation 1, then $A_i^{(n)} = n\, A_i^{(1)} = 0$ for any $n \geq 1$. The script verifies this explicitly for $n = 1, 2, 3$.

7.3 The number three: three-axis hypothesis

[model-dependent] The SM takes the number of generations as an empirical input (the LEP measurement $N_\nu = 2.984 \pm 0.008$ from the Z-boson width [82]). ISPG offers a geometric reading:

- space is three-dimensional;
- a 3D archion admits three orthogonal vibration axes;
- each axis can host one observed family of cavity modes;
- the observed three generations can therefore be mapped to the three axes.

Under this reading, the correspondence

$$\text{Gen-1} \leftrightarrow \hat{x}, \quad \text{Gen-2} \leftrightarrow \hat{y}, \quad \text{Gen-3} \leftrightarrow \hat{z} \tag{9}$$

ties each observed generation to one spatial axis. The exact result in this phase is the quantum-number replication; the three-axis origin of $n_{\text{gen}} = 3$ remains model-dependent until the cavity dynamics also exclude additional family-like modes. In the companion MASS work (§2), the charged-lepton Koide geometry and Sargent-type weak-decay scaling show strong numerical compatibility with this reading, whereas the individual labels $N_\mu = 72$, $N_\tau = 295$, and the ratio $N_\tau/N_e = 59$ remain selected labels whose exact 3D-cavity derivation is still open.

7.4 Fermion count

[derived here] The total number of left-handed Weyl fermion states (counting colors) in three generations is

$$3 \times \big[\, 2 \cdot 1 + 2 \cdot 3 + 1 + 3 + 3 \,\big] \;=\; 3 \times 15 \;=\; 45, \tag{10}$$

matching the SM total exactly. Every one of the 45 states satisfies (7).

7.5 Phase F verdict

The three generations exactly replicate generation 1 at the level of imported SM quantum numbers. If the three-axis hypothesis is adopted, this replica structure can be assigned to the three orthogonal spatial axes. Anomaly cancellation is automatic for any number of replicated generations. Thus Phase F closes the generation-replication bookkeeping, while the claim that spatial dimensionality dynamically fixes $n_{\text{gen}} = 3$ remains a model-dependent open step.

8 Synthesis

8.1 Cross-sector mass-law compatibility

[numerically verified] Applying the large-N form $m_N = (N/N_e)^2\, m_e$ with $N_e = 5$ uniformly across the twelve listed non-neutrino massive Standard-Model states (three charged leptons, six quarks, W, Z, H) yields the following compatibility table. Script: `python/higgs_replacement_unified.py`.

Particle	Sector	N	m_{PDG} (MeV)	m_{ISPG} (MeV)	residual
e	lepton	5	0.5110	0.5110	+0.00%
μ	lepton	72	105.66	105.96	+0.29%
τ	lepton	295	1776.86	1778.79	+0.11%
u	quark	10	2.16	2.044	−5.37%
d	quark	15	4.67	4.599	−1.52%
s	quark	67	93.40	91.76	−1.76%
c	quark	249	1270.0	1267.3	−0.21%
b	quark	452	4180.0	4176.0	−0.10%
t	quark	2906	172 570	172 612	+0.02%
W	gauge	1986	80 369	80 619	+0.31%
Z	gauge	2115	91 188	91 433	+0.27%
H	Higgs	2479	125 250	125 613	+0.29%

Sector-wise absolute residuals: leptons $\leq 0.29\%$, gauge bosons $\leq 0.31\%$, Higgs 0.29%, quarks $\leq 5.37\%$ (current-mass scheme). Overall mean absolute residual is 0.85%; the single outlier is the up-quark current mass (scheme-dependent; the pole/$\overline{\text{MS}}$ discrepancy at this scale is itself of order several percent).

As a further consistency check on the same ladder, the ratio

$$\left(\frac{N_W}{N_Z}\right)^2 = 0.88173 \quad \text{vs.} \quad \left(\frac{m_W}{m_Z}\right)_{\text{PDG}} = 0.88135,$$

i.e. 0.04% agreement with the SM on-shell identity $m_W/m_Z = \cos\theta_W^{\text{os}}$; this is Eq. (4) re-evaluated on the full table. The W mass used here is the PDG 2025 average; the newer CMS 2026 value $m_W = 80\,360.2 \pm 9.9\,\text{MeV}$ does not materially change the compatibility status.

Status of this table: compatibility, not derivation. Every N_f is fit individually to the observed mass through (2); the table is therefore a demonstration that a single integer ladder with a single anchor ($N_e = 5$, m_e) accommodates the twelve listed non-neutrino massive SM states within sub-percent residuals across the heavy sector. Neutrino masses are not included in this ladder table and remain a separate open sector. The structural content that survives parameter counting is the *uniformity*: one formula, one anchor, one ladder, twelve listed particles across four sectors, no Yukawa coupling, no scalar VEV, no sector-specific coupling constant. Promotion from fit to derivation requires producing the N_f themselves from cavity dynamics (items E1 and E5 of §9); the partial 3D-cavity result for $N_e = 5$ and for $\kappa_\tau^2/\kappa_e^2 \simeq N_\tau$ recorded under E5 is the first step of that programme.

8.2 MVP closure

Combining phases A–F:

Phase	Content	Status
A	gauge sector SU(3) × SU(2) × U(1)	SU(3) octet count; E6 algebra/vertices deferred; E
B/C	Gen-1 fermion quantum numbers (Q, T_3, Y, N_c)	imported-SM compatibility
D	EW breaking without Higgs	structural candidate (E7 open)
E	five gauge anomalies	exact imported-hypercharge check
F	three generations as three axes	compatibility (model-dep.)

The ISPG framework therefore *consistently hosts* the SM content at the level checked here: none of the gauge-theoretic requirements of the SM (gauge group dimension, fermion content, anomaly cancellation, m_W/m_Z consistency, generation count) is in contradiction with the ISPG archion + Mathieu-ladder + three-axis framework. A *complete* structural embedding would further require (i) the SU(3) Lie-algebra structure beyond the mode count (E6), (ii) a derived archion-interior mechanism for gauge-boson mass generation (E7), (iii) the individual cavity labels N_W, N_Z (E1), and (iv) a cavity origin for chirality (E2). The present paper reduces the SM-compatibility question to this well-defined list of open items; it does not close them.

8.3 What is derived versus what is imported versus what is fit

- *[derived here]* (structurally, from ISPG substrate): the Mathieu mass law $m \propto N^2$ (from the companion MASS paper); the linearity of gauge anomalies in n_{gen}.
- *[numerically verified]* (partial compatibility clue): the *octet count* of confined modes in the D_3-deformed oscillon toy/robustness analysis; the full $\text{SU}(3)_c$ algebra, vertices, and $G_2(X)$ parameters remain deferred to E6; the checked integer pattern $Y\,N_c \in \mathbb{Z}$ in the imported SM assignment.
- *[numerically verified]*/fit (diagnostic, not predictive): the generation-1 formula (5) (three ingredients vs three data points); the individual cavity labels N_W, N_Z, N_H (fit to m_W, m_Z, m_H); the m_W/m_Z ratio test (self-consistent given the individual fits).
- *[standard SM]* (imported without re-derivation): the gauge-anomaly formulae (1)–(5); the Gell-Mann–Nishijima relation $Q = T_3 + Y/2$; the hypercharge values Y_i (via Gen-1 formula + Gell-Mann–Nishijima); the chirality assignment (left-Weyl basis); the identical (Q, T_3, Y, N_c) across generations.
- *[model-dependent]* (hypothesis): the three-axis reading of the three generations; the archion-interior mechanism for gauge-boson mass.

This separation makes explicit that the "structural embedding" claim rests on items in the first category; items in the second and third categories must be promoted to the first before the embedding becomes a derivation.

8.4 Relation to existing ISPG papers

This article does not introduce any new field, coupling, or action term: the action (1) is unchanged from [45, 47]. The Mathieu ladder is unchanged from the companion MASS paper (`MASS/ISPG_FrequencyToMass.tex`). The eight-gluon derivation is unchanged from the dedicated D_3-oscillon analysis (`SM/ISPG_EightGluon_Robustness.md`). The present paper *synthesizes* these earlier results into a single embedding statement, with the quantum-number maps of §4 and the anomaly analysis of §6 as new content.

9 Extension program

The following questions lie beyond the MVP embedding. We list them not as weaknesses of ISPG but as well-posed structural extensions, each tied to an identifiable computational or theoretical task.

E1. Individual cavity labels N_W, N_Z. The ratio consistency check is already recorded in §3.2; it becomes an independent prediction only if the individual labels N_W and N_Z are derived from the vector-cavity spectrum. That derivation requires diagonalizing the vector Laplacian on the 3D archion background. A plausible attribution (to be tested numerically) is N_W to the $\ell = 1$ triplet with a specific radial quantum number and N_Z to a radial excitation obtained by orthogonalization against a hypercharge-like $\ell = 0$ mode. This is a finite-dimensional numerical task.

E2. Chirality from the cavity. The left/right distinction is currently imported from the Dirac structure of S_m. A cavity origin would require posing the chiral eigenvalue problem on the archion background: identify a parity-breaking boundary condition or a topological invariant that splits the spectrum into left-handed weak doublets and right-handed weak singlets with the imported SM quantum numbers. The success criterion is to recover the SM chirality assignment without postulating it. This is a conceptually distinct task from the embedding itself.

E3. CKM and PMNS matrices. Under the three-axis hypothesis the three generations are assigned to a spatial basis, but the quark and lepton mixing sectors are not the same problem. The small CKM mixing could be represented by perturbative non-orthogonality of the quark axes, while the large PMNS mixing requires a separate dynamical mechanism in the neutrino sector. A first-principles derivation must recover the CKM Wolfenstein parameters, the PMNS angles, the CKM and PMNS CP phases, the neutrino mass ordering, and the Majorana/Dirac character of neutrinos from cavity quantities. At present these are open targets.

E4. α_{NL} and Φ_c. The ISPG Lagrangian contains two dimensionful scales (α_{NL} for the nonlinear self-interaction, Φ_c for the archion amplitude) that are currently fit. A first-principles derivation would proceed via the nonlinear archion profile at fixed cosmological background; recent work identifies this as a BVP in the stationary Φ equation. The notation α_{NL} is used to keep this nonlinear ISPG scale distinct from the fine-structure constant α_{EM}.

E5. Exact mass hierarchy. The charged-lepton hierarchy (m_e, m_μ, m_τ) is numerically reproduced by the three-axis hypothesis to the precisions cited in §2; the quark hierarchy $(m_u \ll m_c \ll m_t)$ is at the same heuristic level. Promoting the three-axis ansatz to a deduction from the archion ground state is the central open task for a mass-exact (as opposed to compatibility-level) embedding.

[numerically verified] A partial structural result is available for the anchor label itself: the electron index $N_e = 5$ matches the $\ell = 2$ degeneracy count of the 3D spherical cavity (five independent spherical harmonics Y_2^m, $m = -2, \ldots, +2$). In the same 3D cavity analysis the ratio N_τ^2/N_e^2 is not reproduced. The actual partial result is $\kappa_\tau^2/\kappa_e^2 \simeq 292.43$, which matches the selected label $N_\tau = 295$ to within 0.9%; equivalently, $\kappa_\tau^2/(5\kappa_e^2) \simeq 58.49$ is close to the selected ratio $N_\tau/N_e = 59$. These results fix part of the kinematic skeleton of the three-axis ansatz (dimensionality of the mode space, radial wavenumber ratios) at the cavity-geometry level, but they do not yet derive the exact integer labels.

[deferred to extension program] Extending the partial result to absorb the 0.9% residual requires a Floquet treatment of the periodic archion: the present analysis uses a peak-amplitude approximation of the nonlinear potential, whereas the correct procedure replaces this by the time-averaged effective potential computed from the full Floquet spectrum at deformation $q \simeq 1.853$. This is a well-defined computational task whose completion would promote E5 from compatibility into derivation.

E6. SU(3) **Lie-algebra structure beyond the count.** Extracting the Gell-Mann commutation relations $[T^a, T^b] = i f^{abc} T^c$ with the standard SU(3) structure constants from the trilinear mode-coupling vertices of the eight confined modes of the D_3-deformed oscillon. Without this step, the gluon count is consistent with several Lie algebras of dimension 8, not only SU(3). *[numerically verified]* A partial group-theoretic result is available: if the three spatial axes are identified with the fundamental **3** of color (as in the three-axis hypothesis used for the lepton sector), the eight generators of unitary mixings among them close on the SU(3) Lie algebra with the textbook structure constants f^{abc} recovered to machine precision ($\leq 2 \times 10^{-16}$ on all nine independent entries, full antisymmetry, and the Jacobi identity; script `python/gluon_su3_closure.py`). This fixes the target algebra but does not replace the dynamical extraction of f^{abc} from the oscillon vertices, which is the content of E6 proper.

E7. Archion-interior gauge-boson mass generation. A derivation showing that an $\mathrm{SU}(2) \times \mathrm{U}(1)$ gauge field minimally coupled to the archion acquires a mass matrix matching (M_W, M_Z) while leaving the photon massless. This is the concrete replacement for the SM Higgs mechanism that the ISPG action forbids; its absence is the single largest dynamical gap in the present paper. The operational target is: write the imported $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge action on the archion background, specify its coupling to $\Phi(r)$, derive the mass matrix in the (W^1, W^2, W^3, B) basis, show that one eigenvalue remains exactly zero (the photon mode), and check the gauge-consistency conditions needed for current conservation and unitarity. Only then can the observed M_W, M_Z, and the E1 cavity labels be treated as derived rather than fit. Plausible route: expand the gauge kinetic term around the radially varying archion profile $\Phi(r)$ and show that the volume integrals yield the observed mass pattern when combined with the cavity labels from E1.

None of these items conflicts with the compatibility established in §§3–7. Each is a well-defined computational or theoretical task whose resolution would promote the present paper from *compatibility check* to *derivation*. Until E1, E6, and E7 are closed, the electroweak and strong sectors should not be represented as "derived" from ISPG but as "consistently hosted in" ISPG.

10 Conclusion

The SM gauge and fermion content is compatible with the ISPG cavity framework at the level checked here: the eight-gluon count matches $\dim \mathrm{SU}(3)_{\mathrm{adj}}$, the Mathieu ladder hosts the electroweak bosons consistently, the SM hypercharges pass all five anomaly tests when imported via the generation-1 fit formula combined with Gell-Mann–Nishijima, and three generations admit a model-dependent geometric reading in terms of three spatial axes. Neutrinos are explicitly excluded from the Mathieu-ladder fermion formula and are handled in the companion MASS paper as polarization-rotation modes.

The embedding is however *partial*: the SU(3) Lie algebra beyond the count (E6), the archion-interior mass mechanism replacing the SM Higgs (E7), the individual cavity labels N_W, N_Z (E1), and the cavity origin of chirality (E2) are all outstanding. The $\cos\theta_W^{\mathrm{os}}$ test of §3.2 is a self-consistency check rather than a derivation.

The value of the present paper lies in having separated these items from one another and from the ISPG substrate: each open item is now a well-posed computational or theoretical task, and none requires modifying the ISPG action (1). Closing them in sequence would convert the present compatibility result into a full derivation of the SM from the single-scalar substrate; the structural path is laid out in §9.

A Python verification scripts

All scripts are self-contained and run from `python/` with no external state.

Script	Purpose	Phase
`phase_a_gauge_sector.py`	EW boson Mathieu indices; on-shell $\cos\theta_W$ ratio test	A
`phase_c_quantum_number_map.py`	Gen-1 formula $N = 5[1 + N_c(1 - \|Q\|)]$ on (e, ν_e, u, d)	C.1
`phase_c_step2_T3_Y_map.py`	Gell-Mann–Nishijima for all 7 L+R Gen-1 fermions; $Y\,N_c$ integrality	C.2
`phase_d_ew_breaking.py`	$m \propto N^2$ test on W, Z, H; absence of ϕ^4 in the action	D
`phase_e_anomalies.py`	Five anomaly conditions, exact rational arithmetic	E
`phase_f_generation_replication.py`	Structural identity of Gen 1,2,3; anomaly linearity in n_{gen}	F
`gluon_su3_closure.py`	Target-algebra compatibility on assumed three colour axes; textbook f^{abc} matched to $\leq 2 \times 10^{-16}$, but no dynamic extraction from confined modes	A compatibil
`higgs_replacement_unified.py`	Unified compatibility table $m = (N/N_e)^2 m_e$ across the 12 listed non-neutrino massive SM states (charged leptons, quarks, W, Z, H); mean residual 0.85%	A–D

Each script prints a human-readable report ending in either a pass line or an explicit failure indicator. The cumulative output constitutes a machine-verifiable audit of the embedding.

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